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LAB REPORT

on

OPERATING SYSTEMS

Submitted by

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in partial fulfillment for the award of the degree of

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CERTIFICATE

This is to certify that the Lab work entitled “OPERATING SYSTEMS – 23CS4PCOPS” carried out by Prerit Jain (1WA24CS218), who is Bonafide student of B. M. S. College of Engineering. It is in partial fulfilment for the award of Bachelor of Engineering in Computer Science and Engineering of the Visvesvaraya Technological University, Belgaum during the year Feb-2026 to June-2026
. The Lab report has been approved as it satisfies the academic requirements in respect of a OPERATING SYSTEMS - (23CS4PCOPS) work prescribed for the said degree.

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Course Outcomes

C01	Apply the different concepts and functionalities of Operating System
C02	Analyse various Operating system strategies and techniques
C03	Demonstrate the different functionalities of Operating System.
C04	Conduct practical experiments to implement the functionalities of Operating system.

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20
24/4

20
8/5

20
22/5

Program 1

Write a C program to simulate the following CPU scheduling algorithm to find turnaround time and waiting time. (Any one)

a) FCFS b) SJF c) Priority d) Round Robin (Experiment with different quantum sizes for RR algorithm)

FCFS:

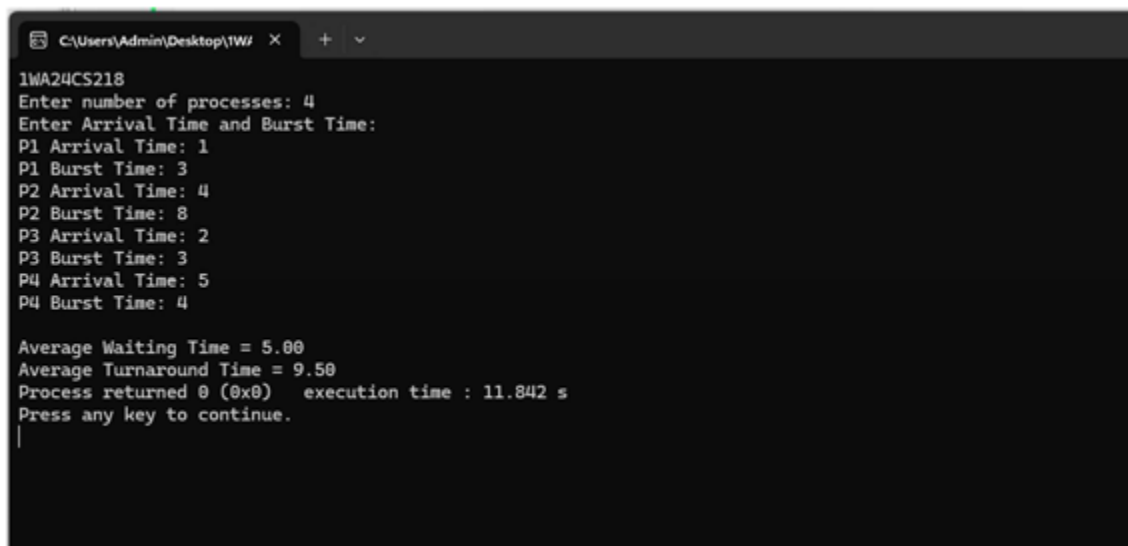
```
#include <stdio.h>

int main() {
    int n, i;
    int at[20], bt[20], wt[20], tat[20], ct[20];
    printf("1WA24CS218\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter Arrival Time and Burst Time:\n");
    for(i = 0; i < n; i++) {
        printf("P%d Arrival Time: ", i+1);
        scanf("%d", &at[i]);
        printf("P%d Burst Time: ", i+1);
        scanf("%d", &bt[i]);
    }
    ct[0] = at[0] + bt[0];
    for(i = 1; i < n; i++) {
        if(ct[i-1] < at[i])
            ct[i] = at[i] + bt[i];
        else
            ct[i] = ct[i-1] + bt[i];
    }
    for(i = 0; i < n; i++)
        tat[i] = ct[i] - at[i];
}
```

```

for(i = 0; i < n; i++)
    wt[i] = tat[i] - bt[i];
float avg_wt = 0, avg_tat = 0;
for(i = 0; i < n; i++) {
    avg_wt += wt[i];
    avg_tat += tat[i];
}
printf("\nAverage Waiting Time = %.2f", avg_wt/n);
printf("\nAverage Turnaround Time = %.2f", avg_tat/n);
return 0;
}

```



```

C:\Users\Admin\Desktop\1W\ x + v
1WA24CS218
Enter number of processes: 4
Enter Arrival Time and Burst Time:
P1 Arrival Time: 1
P1 Burst Time: 3
P2 Arrival Time: 4
P2 Burst Time: 8
P3 Arrival Time: 2
P3 Burst Time: 3
P4 Arrival Time: 5
P4 Burst Time: 4

Average Waiting Time = 5.00
Average Turnaround Time = 9.50
Process returned 0 (0x0)   execution time : 11.842 s
Press any key to continue.
|

```

SJF Non Preemptive:

```
#include <stdio.h>
int main() {
    int n, i, j;
    printf("1WA24CS218\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);
    int at[n], bt[n], ct[n], tat[n], wt[n];
    int finished[n];
    for(i = 0; i < n; i++) {
        printf("Enter Arrival Time and Burst Time for P%d: ", i+1);
        scanf("%d %d", &at[i], &bt[i]);
        finished[i] = 0;
    }
    int current_time = 0;
    int completed = 0;
    while(completed < n) {
        int min_bt = 9999;
        int index = -1;
        for(i = 0; i < n; i++) {
            if(at[i] <= current_time && finished[i] == 0) {
                if(bt[i] < min_bt) {
                    min_bt = bt[i];
                    index = i;
                }
            }
        }
        if(index == -1) {
            current_time++;
        }
        else {
            ct[index] = current_time + bt[index];
            tat[index] = ct[index] - at[index];
            wt[index] = tat[index] - bt[index];
            current_time = ct[index];
            finished[index] = 1;
            completed++;
        }
    }
}
```

```

printf("\nProcess\tAT\tBT\tCT\tTAT\tWT\n");
for(i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1, at[i], bt[i], ct[i], tat[i], wt[i]);
}
return 0;
}

```

1WA24CS218

Enter number of processes: 4

Enter Arrival Time and Burst Time for P1: 0 7

Enter Arrival Time and Burst Time for P2: 8 3

Enter Arrival Time and Burst Time for P3: 3 4

Enter Arrival Time and Burst Time for P4: 5 6

Process	AT	BT	CT	TAT	WT
P1	0	7	7	7	0
P2	8	3	14	6	3
P3	3	4	11	8	4
P4	5	6	20	15	9

Process returned 0 (0x0) execution time : 32.933 s

Press any key to continue.

|

SJF Preemptive:

```
#include <stdio.h>
int main() {
    int n, i;
    printf("1WA24CS18\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);
    int at[n], bt[n], rt[n], ct[n], tat[n], wt[n];
    for(i = 0; i < n; i++) {
        printf("Enter Arrival Time and Burst Time for P%d: ", i+1);
        scanf("%d %d", &at[i], &bt[i]);
        rt[i] = bt[i];
    }
    int current_time = 0, completed = 0;
    while(completed < n) {
        int min_rt = 9999, index = -1;
        for(i = 0; i < n; i++) {
            if(at[i] <= current_time && rt[i] > 0) {
                if(rt[i] < min_rt) {
                    min_rt = rt[i];
                    index = i;
                }
            }
        }
        if(index == -1) {
            current_time++;
        }
        else {
            rt[index]--;
            current_time++;
            if(rt[index] == 0) {
                completed++;
                ct[index] = current_time;
                tat[index] = ct[index] - at[index];
                wt[index] = tat[index] - bt[index];
            }
        }
    }
    printf("\nP\tAT\tBT\tCT\tTAT\tWT\n");
```

```

for(i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1, at[i], bt[i], ct[i], tat[i], wt[i]);
}
return 0;
}

```

```

1WA24CS18
Enter number of processes: 4
Enter Arrival Time and Burst Time for P1: 0 8
Enter Arrival Time and Burst Time for P2: 1 4
Enter Arrival Time and Burst Time for P3: 2 9
Enter Arrival Time and Burst Time for P4: 3 5

P      AT      BT      CT      TAT      WT
P1      0      8      17      17      9
P2      1      4      5       4      0
P3      2      9      26      24     15
P4      3      5      10      7       2

Process returned 0 (0x0)   execution time : 29.528 s
Press any key to continue.
|

```

Priority Non Preemptive:

```
#include <stdio.h>
int main() {
    int n, i;
    int pid[20], at[20], bt[20], pr[20];
    int wt[20], tat[20], ct[20];
    int completed[20] = {0};
    int current_time = 0, completed_count = 0;
    int highest_priority, selected;
    float avg_wt = 0, avg_tat = 0;
    printf("1WA24CS218\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);
    for(i = 0; i < n; i++) {
        pid[i] = i + 1;
        printf("P%d Arrival Time: ", i+1);
        scanf("%d", &at[i]);
        printf("P%d Burst Time: ", i+1);
        scanf("%d", &bt[i]);
        printf("P%d Priority: ", i+1);
        scanf("%d", &pr[i]);
    }
    while(completed_count < n) {
        highest_priority = 999;
        selected = -1;
        for(i = 0; i < n; i++) {
            if(at[i] <= current_time && completed[i] == 0) {
                if(pr[i] < highest_priority) {
                    highest_priority = pr[i];
                    selected = i;
                }
            }
        }
        if(selected == -1) {
            current_time++;
        }
        else {
            ct[selected] = current_time + bt[selected];
            tat[selected] = ct[selected] - at[selected];
        }
    }
}
```

```

        wt[selected] = tat[selected] - bt[selected];
        current_time = ct[selected];
        completed[selected] = 1;
        completed_count++;
        avg_wt += wt[selected];
        avg_tat += tat[selected];
    }
}
printf("\nProcess\tAT\tBT\tPR\tCT\tTAT\tWT\n");
for(i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        pid[i], at[i], bt[i], pr[i], ct[i], tat[i], wt[i]);
}
avg_wt /= n;
avg_tat /= n;
printf("\nAverage Waiting Time = %.2f", avg_wt);
printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
return 0;
}

```

```

1WA24CS218
Enter number of processes: 5
P1 Arrival Time: 0
P1 Burst Time: 3
P1 Priority: 5
P2 Arrival Time: 2
P2 Burst Time: 2
P2 Priority: 3
P3 Arrival Time: 3
P3 Burst Time: 5
P3 Priority: 2
P4 Arrival Time: 4
P4 Burst Time: 4
P4 Priority: 4
P5 Arrival Time: 6
P5 Burst Time: 1
P5 Priority: 1

Process AT    BT    PR    CT    TAT    WT
P1      0      3      5      3      3      0
P2      2      2      3     11      9      7
P3      3      5      2      8      5      0
P4      4      4      4     15     11      7
P5      6      1      1      9      3      2

Average Waiting Time = 3.20
Average Turnaround Time = 6.20

Process returned 0 (0x0)   execution time : 59.375 s
Press any key to continue.

```

Priority Preemptive:

```
#include <stdio.h>
int main() {
    int n, i, time = 0, completed = 0;
    int pid[20], at[20], bt[20], pr[20];
    int rt[20], ct[20], tat[20], wt[20];
    int highest, pos;
    float avg_wt = 0, avg_tat = 0;
    printf("1WA24CS218\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);
    for(i = 0; i < n; i++) {
        pid[i] = i + 1;
        printf("P%d Arrival Time: ", i+1);
        scanf("%d", &at[i]);
        printf("P%d Burst Time: ", i+1);
        scanf("%d", &bt[i]);
        printf("P%d Priority: ", i+1);
        scanf("%d", &pr[i]);
        rt[i] = bt[i];
    }
    while(completed < n) {
        highest = 999;
        pos = -1;
        for(i = 0; i < n; i++) {
            if(at[i] <= time && rt[i] > 0 && pr[i] < highest) {
                highest = pr[i];
                pos = i;
            }
        }
        if(pos == -1) {
            time++;
        }
        else {
            rt[pos]--;
            time++;
            if(rt[pos] == 0) {
                completed++;
                ct[pos] = time;
            }
        }
    }
}
```

```

        tat[pos] = ct[pos] - at[pos];
        wt[pos] = tat[pos] - bt[pos];
        avg_wt += wt[pos];
        avg_tat += tat[pos];
    }
}
}
printf("\nProcess\tAT\tBT\tPR\tCT\tTAT\tWT\n");
for(i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
        pid[i], at[i], bt[i], pr[i], ct[i], tat[i], wt[i]);
}
avg_wt /= n;
avg_tat /= n;
printf("\nAverage Waiting Time = %.2f", avg_wt);
printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
return 0;
}

```

```

1WA24CS218
Enter number of processes: 5
P1 Arrival Time: 0
P1 Burst Time: 3
P1 Priority: 3
P2 Arrival Time: 1
P2 Burst Time: 4
P2 Priority: 2
P3 Arrival Time: 2
P3 Burst Time: 6
P3 Priority: 4
P4 Arrival Time: 3
P4 Burst Time: 4
P4 Priority: 6
P5 Arrival Time: 4
P5 Burst Time: 2
P5 Priority: 10

Process AT      BT      PR      CT      TAT      WT
P1      0       3       3       7       7       4
P2      1       4       2       5       4       0
P3      2       6       4      13      11       5
P4      3       4       6      17      14      10
P5      4       2      10      19      15      13

Average Waiting Time = 6.40
Average Turnaround Time = 10.20

Process returned 0 (0x0)   execution time : 22.164 s
Press any key to continue.

```

Round Robin:

```
#include <stdio.h>
int main() {
    int n, quantum, time = 0, completed = 0;
    int queue[100], front = 0, rear = 0;
    int visited[100] = {0};
    printf("IWA24CS218\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);
    int at[n], bt[n], rt[n], ct[n], tat[n], wt[n];
    for (int i = 0; i < n; i++) {
        printf("Enter AT and BT for P%d: ", i + 1);
        scanf("%d %d", &at[i], &bt[i]);
        rt[i] = bt[i];
    }
    printf("Enter Time Quantum: ");
    scanf("%d", &quantum);
    for(int i = 0; i < n; i++) {
        if(at[i] <= time) {
            queue[rear++] = i;
            visited[i] = 1;
        }
    }
    while (front < rear) {
        int i = queue[front++];
        int execute = (rt[i] > quantum) ? quantum : rt[i];
        rt[i] -= execute;
        time += execute;
        for (int j = 0; j < n; j++) {
            if (at[j] <= time && visited[j] == 0) {
                queue[rear++] = j;
                visited[j] = 1;
            }
        }
        if (rt[i] > 0) {
            queue[rear++] = i;
        }
        else {
            ct[i] = time;
        }
    }
}
```

```

        completed++;
    }
    if (front == rear && completed < n) {
        for (int j = 0; j < n; j++) {
            if (visited[j] == 0) {
                time = at[j];
                queue[rear++] = j;
                visited[j] = 1;
                break;
            }
        }
    }
}
printf("\nID\tAT\tBT\tCT\tTAT\tWT\n");
for (int i = 0; i < n; i++) {
    printf("%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1, at[i], bt[i], ct[i],
        ct[i]-at[i],
        (ct[i]-at[i])-bt[i]);
}
return 0;
}

```

```

1WA24CS218
Enter number of processes: 5
Enter AT and BT for P1: 0 5
Enter AT and BT for P2: 1 3
Enter AT and BT for P3: 2 1
Enter AT and BT for P4: 3 2
Enter AT and BT for P5: 4 3
Enter Time Quantum: 2

```

ID	AT	BT	CT	TAT	WT
1	0	5	13	13	8
2	1	3	12	11	8
3	2	1	5	3	2
4	3	2	9	6	4
5	4	3	14	10	7

```

Process returned 0 (0x0)   execution time : 26.236 s
Press any key to continue.
|

```


Program 2

Write a C program to simulate multi-level queue scheduling algorithm considering the following scenario. All the processes in the system are divided into two categories – system processes and user processes. System processes are to be given higher priority than user processes. Use FCFS scheduling for the processes in each queue.

```
#include <stdio.h>
#include <string.h>
#define MAX 100
struct Process {
    char pid[5];
    int arrival, burst, remaining;
    int completion, turnaround, waiting;
    int qno;
    int done;
};
int main() {
    struct Process p[MAX];
    int n, i, completed = 0;
    int current_time = 0, tq;
    float total_tat = 0, total_wt = 0;
    printf("1WA24CS218\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);
    for(i = 0; i < n; i++) {
        printf(p[i].pid, "p%d", i+1);
        printf("\nProcess %s\n", p[i].pid);
        printf("Enter Arrival Time: ");
        scanf("%d", &p[i].arrival);
        printf("Enter Burst Time: ");
        scanf("%d", &p[i].burst);
        p[i].remaining = p[i].burst;
        p[i].done = 0;
        printf("Enter Queue (1-System, 2-User): ");
        scanf("%d", &p[i].qno);
    }
    printf("\nEnter Time Quantum for System Queue: ");
    scanf("%d", &tq);
    while(completed < n) {
        int executed = 0;
```

```

for(i = 0; i < n; i++) {
    if(p[i].qno == 1 && p[i].arrival <= current_time && p[i].remaining > 0) {
        executed = 1;
        if(p[i].remaining > tq) {
            p[i].remaining -= tq;
            current_time += tq;
        }
        else {
            current_time += p[i].remaining;
            p[i].remaining = 0;
            p[i].completion = current_time;
            p[i].turnaround = p[i].completion - p[i].arrival;
            p[i].waiting = p[i].turnaround - p[i].burst;
            total_tat += p[i].turnaround;
            total_wt += p[i].waiting;
            completed++;
        }
    }
}
if(executed)
    continue;
int idx = -1;
int min_arrival = 1e9;
for(i = 0; i < n; i++) {
    if(p[i].qno == 2 && p[i].arrival <= current_time && p[i].remaining > 0) {
        if(p[i].arrival < min_arrival) {
            min_arrival = p[i].arrival;
            idx = i;
        }
    }
}
if(idx != -1) {
    executed = 1;
    int run_time = p[idx].remaining;
    for(int t = 1; t <= run_time; t++) {
        current_time++;
        p[idx].remaining--;
        int sys_arrived = 0;
        for(i = 0; i < n; i++) {
            if(p[i].qno == 1 && p[i].arrival <= current_time && p[i].remaining > 0) {

```

```

        sys_arrived = 1;
        break;
    }
}
if(sys_arrived)
    break;
}
if(p[idx].remaining == 0) {
    p[idx].completion = current_time;
    p[idx].turnaround = p[idx].completion - p[idx].arrival;
    p[idx].waiting = p[idx].turnaround - p[idx].burst;
    total_tat += p[idx].turnaround;
    total_wt += p[idx].waiting;
    completed++;
}
}
if(!executed)
    current_time++;
}
printf("\nPID\tQ\tAT\tBT\tCT\tTAT\tWT\n");
for(i = 0; i < n; i++) {
    printf("%s\t%d\t%d\t%d\t%d\t%d\t%d\n",
        p[i].pid, p[i].qno, p[i].arrival, p[i].burst,
        p[i].completion, p[i].turnaround, p[i].waiting);
}
printf("\nAverage Turnaround Time = %.2f", total_tat / n);
printf("\nAverage Waiting Time = %.2f\n", total_wt / n);
return 0;
}

```

1WA24CS218

Enter number of processes: 4

Process p1

Enter Arrival Time: 0

Enter Burst Time: 4

Enter Queue (1-System, 2-User): 1

Process p2

Enter Arrival Time: 0

Enter Burst Time: 3

Enter Queue (1-System, 2-User): 1

Process p3

Enter Arrival Time: 0

Enter Burst Time: 8

Enter Queue (1-System, 2-User): 2

Process p4

Enter Arrival Time: 10

Enter Burst Time: 5

Enter Queue (1-System, 2-User): 1

Enter Time Quantum for System Queue: 2

PID	Q	AT	BT	CT	TAT	WT
p1	1	0	4	6	6	2
p2	1	0	3	7	7	4
p3	2	0	8	20	20	12
p4	1	10	5	15	5	0

Average Turnaround Time = 9.50

Average Waiting Time = 4.50

Program 3

Write a C program to simulate Real-Time CPU Scheduling algorithms: (Any one)

a) Rate- Monotonic b) Earliest-deadline First c) Proportional scheduling

Rate Monotonic:

```
#include <stdio.h>
int gcd(int a, int b) {
    while (b) {
        int t = b;
        b = a % b;
        a = t;
    }
    return a;
}
int lcm(int a, int b) {
    return (a * b) / gcd(a, b);
}
struct Task {
    int id;
    int burst;
    int period;
    int remaining;
};
int main() {
    int n;
    printf("1WA24CS218\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);
    struct Task t[n];
    printf("Enter burst times:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &t[i].burst);
        t[i].id = i + 1;
    }
    printf("Enter periods:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &t[i].period);
        t[i].remaining = 0;
    }
}
```

```

for (int i = 0; i < n - 1; i++) {
    for (int j = i + 1; j < n; j++) {
        if (t[i].period > t[j].period) {
            struct Task temp = t[i];
            t[i] = t[j];
            t[j] = temp;
        }
    }
}
int hyper = t[0].period;
for (int i = 1; i < n; i++)
    hyper = lcm(hyper, t[i].period);
float util = 0;
for (int i = 0; i < n; i++)
    util += (float)t[i].burst / t[i].period;
float bound;
if (n == 1)
    bound = 1.0;
else if (n == 2)
    bound = 0.828427;
else if (n == 3)
    bound = 0.779763;
else
    bound = 0.693;
printf("\n%f <= %f => %s\n", util, bound, (util <= bound) ? "true" : "false");
printf("\nScheduling occurs for %d ms\n\n", hyper);
int remaining[n];
for (int i = 0; i < n; i++)
    remaining[i] = 0;
int last = -2;
for (int time = 0; time < hyper; time++) {
    for (int i = 0; i < n; i++) {
        if (time % t[i].period == 0)
            remaining[i] = t[i].burst;
    }
    int run = -1;
    for (int i = 0; i < n; i++) {
        if (remaining[i] > 0) {
            run = i;
            break;
        }
    }
}

```

```

    }
}
if (run != last) {
    if (run != -1)
        printf("%dms onwards: Process %d running\n", time, t[run].id);
    else
        printf("%dms onwards: CPU is idle\n", time);
}
if (run != -1)
    remaining[run]--;
last = run;
}
return 0;
}

```

```

1WA24CS218
Enter number of processes: 3
Enter burst times:
3 2 2
Enter periods:
20 5 10

0.750000 <= 0.779763 => true

Scheduling occurs for 20 ms

0ms onwards: Process 2 running
2ms onwards: Process 3 running
4ms onwards: Process 1 running
5ms onwards: Process 2 running
7ms onwards: Process 1 running
9ms onwards: CPU is idle
10ms onwards: Process 2 running
12ms onwards: Process 3 running
14ms onwards: CPU is idle
15ms onwards: Process 2 running
17ms onwards: CPU is idle

Process returned 0 (0x0)   execution time : 10.253 s
Press any key to continue.
|

```

Earliest Deadline First:

```
#include <stdio.h>
#include <limits.h>
typedef struct {
    int id;
    int burst;
    int deadline;
    int period;
    int remaining_burst;
    int next_deadline;
} Task;
int main() {
    int n, total_time;
    printf("1WA24CS218\n");
    printf("Enter number of tasks: ");
    scanf("%d", &n);
    Task tasks[n];
    for (int i = 0; i < n; i++) {
        printf("Enter PID, Burst, Deadline, and Period for Task %d: ", i + 1);
        scanf("%d %d %d %d",
            &tasks[i].id,
            &tasks[i].burst,
            &tasks[i].deadline,
            &tasks[i].period);
        tasks[i].remaining_burst = 0;
        tasks[i].next_deadline = 0;
    }
    printf("Enter total scheduling time (ms): ");
    scanf("%d", &total_time);
    printf("\nScheduling occurs for %d ms\n\n", total_time);
    for (int t = 0; t < total_time; t++) {
        for (int i = 0; i < n; i++) {
            if (t % tasks[i].period == 0) {
                tasks[i].remaining_burst = tasks[i].burst;
                tasks[i].next_deadline = t + tasks[i].deadline;
            }
        }
    }
    int selected = -1;
    int min_deadline = INT_MAX;
```



```

for (int i = 0; i < n; i++) {
    if (tasks[i].remaining_burst > 0) {
        if (tasks[i].next_deadline < min_deadline) {
            min_deadline = tasks[i].next_deadline;
            selected = i;
        }
        else if (tasks[i].next_deadline == min_deadline) {
            if (selected == -1 || tasks[i].id < tasks[selected].id) {
                selected = i;
            }
        }
    }
}
if (selected == -1) {
    printf("%dms : CPU is idle.\n", t);
}
else {
    printf("%dms : Task %d is running.\n", t, tasks[selected].id);
    tasks[selected].remaining_burst--;
}
}
return 0;
}

```

```

1WA24CS218
Enter number of tasks: 3
Enter PID, Burst, Deadline, and Period for Task 1: 1 3 7 20
Enter PID, Burst, Deadline, and Period for Task 2: 2 2 4 5
Enter PID, Burst, Deadline, and Period for Task 3: 3 2 8 10
Enter total scheduling time (ms): 20

Scheduling occurs for 20 ms

0ms : Task 2 is running.
1ms : Task 2 is running.
2ms : Task 1 is running.
3ms : Task 1 is running.
4ms : Task 1 is running.
5ms : Task 3 is running.
6ms : Task 3 is running.
7ms : Task 2 is running.
8ms : Task 2 is running.
9ms : CPU is idle.
10ms : Task 2 is running.
11ms : Task 2 is running.
12ms : Task 3 is running.
13ms : Task 3 is running.
14ms : CPU is idle.
15ms : Task 2 is running.
16ms : Task 2 is running.
17ms : CPU is idle.
18ms : CPU is idle.
19ms : CPU is idle.

Process returned 0 (0x0)   execution time : 25.097 s
Press any key to continue.
|

```

Proportional Scheduling:

```
#include<stdio.h>
#include <stdlib.h>
#include <time.h>
typedef struct {
    char name[5];
    int tickets;
} Process;
int main() {
    int n, total_tickets = 0;
    float total_T = 0.0;
    printf("1WA24CS218\n");
    printf("Enter the number of Processes: ");
    scanf("%d", &n);
    Process p[n];
    srand(time(NULL));
    for (int i = 0; i < n; i++) {
        printf("\nProcess %d:\n", i + 1);
        sprintf(p[i].name, "P%d", i + 1);
        printf("Tickets: ");
        scanf("%d", &p[i].tickets);
        total_tickets += p[i].tickets;
        total_T += p[i].tickets;
    }
    printf("\n--- Proportional Share Scheduling ---\n");
    printf("Enter the Time Period for scheduling: ");
    int m;
    scanf("%d", &m);
    for (int i = 0; i < m; i++) {
        int winning_ticket = rand() % total_tickets + 1;
        int accumulated_tickets = 0;
        int winner_index;
        for (int j = 0; j < n; j++) {
            accumulated_tickets += p[j].tickets;
            if (winning_ticket <= accumulated_tickets) {
                winner_index = j;
                break;
            }
        }
    }
}
```

```

    printf("Tickets picked: %d, Winner: %s\n",
           winning_ticket,
           p[winner_index].name);
}
for (int i = 0; i < n; i++) {
    printf("\nThe Process: %s gets %0.2f%% of Processor Time.\n",
           p[i].name,
           ((p[i].tickets / total_T) * 100));
}
return 0;
}

```

```

1WA24CS218
Enter the number of Processes: 3

Process 1:
Tickets: 10

Process 2:
Tickets: 20

Process 3:
Tickets: 30

--- Proportional Share Scheduling ---
Enter the Time Period for scheduling: 5
Tickets picked: 44, Winner: P3
Tickets picked: 31, Winner: P3
Tickets picked: 41, Winner: P3
Tickets picked: 42, Winner: P3
Tickets picked: 5, Winner: P1

The Process: P1 gets 16.67% of Processor Time.

The Process: P2 gets 33.33% of Processor Time.

The Process: P3 gets 50.00% of Processor Time.

Process returned 0 (0x0)   execution time : 15.114 s
Press any key to continue.
|

```

Program 4

Write a C program to simulate: (Any one)

a) Producer-Consumer problem using semaphores. b) Dining-Philosopher's problem

Producer Consumer:

```
#include <stdio.h>
#include <stdlib.h>
#define BUFFER_SIZE 5
int buffer[BUFFER_SIZE];
int in = 0, out = 0;
int mutex = 1;
int empty = BUFFER_SIZE;
int full = 0;
int item = 0;
void wait(int *s) {
    (*s)--;
}
void signal(int *s) {
    (*s)++;
}
void produce() {
    if (empty == 0) {
        printf("Buffer is full\n");
        return;
    }
    wait(&empty);
    wait(&mutex);
    item++;
    buffer[in] = item;
    in = (in + 1) % BUFFER_SIZE;
    printf("Producer has produced: item%d\n", item);
    signal(&mutex);
    signal(&full);
}
void consume() {
    if (full == 0) {
        printf("Buffer is empty\n");
        return;
    }
}
```

```

wait(&full);
wait(&mutex);
int consumed = buffer[out];
out = (out + 1) % BUFFER_SIZE;
printf("Consumer has consumed: item%d\n", consumed);
signal(&mutex);
signal(&empty);
}
int main() {
    int choice;
    printf("\n1 WA24CS218");
    while (1) {
        printf("\nEnter your choice:\n");
        printf("1. Producer\n2. Consumer\n3. Exit\n");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                produce();
                break;
            case 2:
                consume();
                break;
            case 3:
                exit(0);
            default:
                printf("Invalid choice\n");
        }
    }
    return 0;
}

```

```
1WA24CS218
Enter your choice:
1. Producer
2. Consumer
3. Exit
1
Producer has produced: item1

Enter your choice:
1. Producer
2. Consumer
3. Exit
1
Producer has produced: item2

Enter your choice:
1. Producer
2. Consumer
3. Exit
2
Consumer has consumed: item1

Enter your choice:
1. Producer
2. Consumer
3. Exit
2
Consumer has consumed: item2

Enter your choice:
1. Producer
2. Consumer
3. Exit
3

Process returned 0 (0x0)   execution time : 11.373 s
Press any key to continue.
|
```

Dining Philosopher Problem:

```
#include <stdio.h>
#include <pthread.h>
#include <unistd.h>
#define N 5
pthread_mutex_t forks[N];
pthread_t philosophers[N];
void* philosopher(void* num)
{
    int id = *(int*)num;
    int left = id;
    int right = (id + 1) % N;
    while (1)
    {
        printf("Philosopher %d is thinking.\n", id);
        sleep(1);
        if (id % 2 == 0)
        {
            pthread_mutex_lock(&forks[left]);
            printf("Philosopher %d picked left fork %d\n", id, left);
            pthread_mutex_lock(&forks[right]);
            printf("Philosopher %d picked right fork %d\n", id, right);
        }
        else
        {
            pthread_mutex_lock(&forks[right]);
            printf("Philosopher %d picked right fork %d\n", id, right);
            pthread_mutex_lock(&forks[left]);
            printf("Philosopher %d picked left fork %d\n", id, left);
        }
        printf("Philosopher %d is eating.\n", id);
        sleep(2);
        pthread_mutex_unlock(&forks[left]);
        pthread_mutex_unlock(&forks[right]);
        printf("Philosopher %d put down forks %d and %d\n", id, left, right);
    }
    return NULL;
}
```

```

int main()
{
    int i;
    int ids[N];
    for (i = 0; i < N; i++)
    {
        pthread_mutex_init(&forks[i], NULL);
        ids[i] = i;
    }
    for (i = 0; i < N; i++)
    {
        pthread_create(&philosophers[i], NULL, philosopher, &ids[i]);
    }
    for (i = 0; i < N; i++)
    {
        pthread_join(philosophers[i], NULL);
    }
    for (i = 0; i < N; i++)
    {
        pthread_mutex_destroy(&forks[i]);
    }
    return 0;
}

```

```

Philosopher 1 is thinking.
Philosopher 0 is thinking.
Philosopher 4 is thinking.
Philosopher 2 is thinking.
Philosopher 3 is thinking.
Philosopher 3 picked right fork 4
Philosopher 3 picked left fork 3
Philosopher 3 is eating.
Philosopher 2 picked left fork 2
Philosopher 0 picked left fork 0
Philosopher 0 picked right fork 1
Philosopher 0 is eating.
Philosopher 0 put down forks 0 and 1
Philosopher 0 is thinking.
Philosopher 2 picked right fork 3
Philosopher 2 is eating.
Philosopher 3 put down forks 3 and 4
Philosopher 3 is thinking.
Philosopher 4 picked left fork 4
Philosopher 4 picked right fork 0
Philosopher 4 is eating.
Philosopher 2 put down forks 2 and 3
Philosopher 4 put down forks 4 and 0
Philosopher 4 is thinking.
Philosopher 0 picked left fork 0

```


Program 5

Write a C program to simulate: (Any one)

a) Bankers' algorithm for the purpose of deadlock avoidance. b) Deadlock Detection

Banker's Algorithm:

```
#include <stdio.h>
int main() {
    int n, m, i, j, k;
    printf("1WA24CS218\n");
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter number of resources: ");
    scanf("%d", &m);
    int allocation[n][m];
    int max[n][m];
    int need[n][m];
    int available[m];
    int work[m];
    int finish[n];
    int safeSequence[n];
    printf("\nEnter Allocation Matrix:\n");
    for(i = 0; i < n; i++) {
        for(j = 0; j < m; j++) {
            scanf("%d", &allocation[i][j]);
        }
    }
    printf("\nEnter Maximum Demand Matrix:\n");
    for(i = 0; i < n; i++) {
        for(j = 0; j < m; j++) {
            scanf("%d", &max[i][j]);
        }
    }
    printf("\nEnter Available Resources:\n");
    for(i = 0; i < m; i++) {
        scanf("%d", &available[i]);
        work[i] = available[i];
    }
    for(i = 0; i < n; i++) {
        for(j = 0; j < m; j++) {
```

```

        need[i][j] = max[i][j] - allocation[i][j];
    }
}
for(i = 0; i < n; i++) {
    finish[i] = 0;
}
int count = 0;
while(count < n) {
    int found = 0;
    for(i = 0; i < n; i++) {
        if(finish[i] == 0) {
            int possible = 1;
            for(j = 0; j < m; j++) {
                if(need[i][j] > work[j]) {
                    possible = 0;
                    break;
                }
            }
            if(possible) {
                for(k = 0; k < m; k++) {
                    work[k] += allocation[i][k];
                }
                safeSequence[count] = i;
                count++;
                finish[i] = 1;
                found = 1;
            }
        }
    }
    if(found == 0)
        break;
}
int safe = 1;
for(i = 0; i < n; i++) {
    if(finish[i] == 0) {
        safe = 0;
        break;
    }
}
if(safe) {

```

```

printf("\nSystem is in a safe state.\n");
printf("Safe sequence is: ");
for(i = 0; i < n; i++) {
    printf("P%d", safeSequence[i]);
    if(i != n - 1)
        printf(" -> ");
}
printf("\n");
}
else {
    printf("\nSystem is not in a safe state.\n");
}
return 0;
}

```

```

1WA24CS218
Enter number of processes: 5
Enter number of resources: 3

Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2

Enter Maximum Demand Matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3

Enter Available Resources:
3 3 2

System is in a safe state.
Safe sequence is: P1 -> P3 -> P4 -> P0 -> P2

Process returned 0 (0x0)   execution time : 67.699 s
Press any key to continue.
|

```

Deadlock Detection:

```
#include <stdio.h>
int main() {
    int n, m, i, j, k;
    printf("1WA24CS218\n");
    printf("Enter the number of processes: ");
    scanf("%d", &n);
    printf("Enter the number of resources: ");
    scanf("%d", &m);
    int allocation[n][m];
    int request[n][m];
    int available[m];
    int work[m];
    int finish[n];
    int safeSequence[n];
    printf("Enter the allocation matrix:\n");
    for(i = 0; i < n; i++) {
        printf("Process %d: ", i);
        for(j = 0; j < m; j++) {
            scanf("%d", &allocation[i][j]);
        }
    }
    printf("Enter the request matrix:\n");
    for(i = 0; i < n; i++) {
        printf("Process %d: ", i);
        for(j = 0; j < m; j++) {
            scanf("%d", &request[i][j]);
        }
    }
    printf("Enter the available resources: ");
    for(i = 0; i < m; i++) {
        scanf("%d", &available[i]);
        work[i] = available[i];
    }
    for(i = 0; i < n; i++) {
        int allZero = 1;
        for(j = 0; j < m; j++) {
            if(allocation[i][j] != 0) {
                allZero = 0;
            }
        }
    }
}
```

```

        break;
    }
}
if(allZero)
    finish[i] = 1;
else
    finish[i] = 0;
}
int count = 0;
while(count < n) {
    int found = 0;
    for(i = 0; i < n; i++) {
        if(finish[i] == 0) {
            int possible = 1;
            for(j = 0; j < m; j++) {
                if(request[i][j] > work[j]) {
                    possible = 0;
                    break;
                }
            }
            if(possible) {
                for(k = 0; k < m; k++) {
                    work[k] += allocation[i][k];
                }
                safeSequence[count] = i;
                count++;
                finish[i] = 1;
                found = 1;
            }
        }
    }
    if(found == 0)
        break;
}
int deadlock = 0;
for(i = 0; i < n; i++) {
    if(finish[i] == 0) {
        deadlock = 1;
        break;
    }
}

```

```

}
if(deadlock) {
    printf("\nSystem is in DEADLOCK state.\n");
    printf("Deadlocked processes are: ");
    for(i = 0; i < n; i++) {
        if(finish[i] == 0)
            printf("P%d ", i);
    }
    printf("\n");
}
else {
    printf("\nSystem is in safe state.\n");
    printf("Safe Sequence is: ");
    for(i = 0; i < n; i++) {
        printf("P%d ", safeSequence[i]);
    }
    printf("\n");
}
return 0;
}

```

```

1WA24CS218
Enter the number of processes: 5
Enter the number of resources: 3
Enter the allocation matrix:
Process 0: 0 1 0
Process 1: 2 0 0
Process 2: 3 0 3
Process 3: 2 1 1
Process 4: 0 0 2
Enter the request matrix:
Process 0: 0 0 0
Process 1: 2 0 2
Process 2: 0 0 0
Process 3: 1 0 0
Process 4: 0 0 2
Enter the available resources: 0 0 0

System is in safe state.
Safe Sequence is: P0 P2 P3 P4 P1

Process returned 0 (0x0)   execution time : 36.712 s
Press any key to continue.
|

```

Program 6

Write a C program to simulate the following contiguous memory allocation techniques. (Any one)

a) Worst-fit b) Best-fit c) First-fit

```
#include <stdio.h>

void firstFit(int blockSize[], int m, int processSize[], int n)
{
    int allocation[50];
    int used[50] = {0};
    for(int i = 0; i < n; i++)
        allocation[i] = -1;
    for(int i = 0; i < n; i++)
    {
        for(int j = 0; j < m; j++)
        {
            if(blockSize[j] >= processSize[i] && used[j] == 0)
            {
                allocation[i] = j;
                used[j] = 1;
                break;
            }
        }
    }
    printf("\n--- First Fit ---\n");
    printf("Process No.\tProcess Size\tBlock No.\n");
    for(int i = 0; i < n; i++)
    {
        printf("%d\t%d\t", i + 1, processSize[i]);
        if(allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }
}

void bestFit(int blockSize[], int m, int processSize[], int n)
{
    int allocation[50];
    int used[50] = {0};
    for(int i = 0; i < n; i++)
```

```

        allocation[i] = -1;
    for(int i = 0; i < n; i++)
    {
        int bestIdx = -1;
        for(int j = 0; j < m; j++)
        {
            if(blockSize[j] >= processSize[i] && used[j] == 0)
            {
                if(bestIdx == -1 || blockSize[j] < blockSize[bestIdx])
                    bestIdx = j;
            }
        }
        if(bestIdx != -1)
        {
            allocation[i] = bestIdx;
            used[bestIdx] = 1;
        }
    }
    printf("\n--- Best Fit ---\n");
    printf("Process No.\tProcess Size\tBlock No.\n");
    for(int i = 0; i < n; i++)
    {
        printf("%d\t%d\t", i + 1, processSize[i]);
        if(allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }
}

void worstFit(int blockSize[], int m, int processSize[], int n)
{
    int allocation[50];
    int used[50] = {0};
    for(int i = 0; i < n; i++)
        allocation[i] = -1;
    for(int i = 0; i < n; i++)
    {
        int worstIdx = -1;
        for(int j = 0; j < m; j++)
        {

```



```

        if(blockSize[j] >= processSize[i] && used[j] == 0)
        {
            if(worstIdx == -1 || blockSize[j] > blockSize[worstIdx])
                worstIdx = j;
        }
    }
    if(worstIdx != -1)
    {
        allocation[i] = worstIdx;
        used[worstIdx] = 1;
    }
}
printf("\n--- Worst Fit ---\n");
printf("Process No.\tProcess Size\tBlock No.\n");
for(int i = 0; i < n; i++)
{
    printf("%d\t\t%d\t\t", i + 1, processSize[i]);
    if(allocation[i] != -1)
        printf("%d\n", allocation[i] + 1);
    else
        printf("Not Allocated\n");
}
}
int main()
{
    int m, n;
    int blockSize[50], processSize[50];
    printf("1WA24CS218\n");
    printf("Enter number of memory blocks: ");
    scanf("%d", &m);
    printf("Enter sizes of %d memory blocks:\n", m);
    for(int i = 0; i < m; i++)
        scanf("%d", &blockSize[i]);
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter sizes of %d processes:\n", n);
    for(int i = 0; i < n; i++)
        scanf("%d", &processSize[i]);
    firstFit(blockSize, m, processSize, n);
    bestFit(blockSize, m, processSize, n);
}

```

```

    worstFit(blockSize, m, processSize, n);
    return 0;
}

```

```

1WA24CS218
Enter number of memory blocks: 5
Enter sizes of 5 memory blocks:
100 500 200 300 600
Enter number of processes: 4
Enter sizes of 4 processes:
212 417 112 426

--- First Fit ---
Process No.      Process Size      Block No.
1                212                2
2                417                5
3                112                3
4                426                Not Allocated

--- Best Fit ---
Process No.      Process Size      Block No.
1                212                4
2                417                2
3                112                3
4                426                5

--- Worst Fit ---
Process No.      Process Size      Block No.
1                212                5
2                417                2
3                112                4
4                426                Not Allocated

```

Program 7

Write a C program to simulate page replacement algorithms. (Any one)

a) FIFO b) LRU c) Optimal

```
#include <stdio.h>
#include <stdlib.h>
void simulate(int pages[], int n, int nf, int mode) {
    int frames[10] = {0}, time[10] = {0}, faults = 0, next = 0;
    for (int i = 0; i < nf; i++)
        frames[i] = -1;
    printf("\n--- %s Page Replacement ---\n",
        mode == 0 ? "FIFO" : mode == 1 ? "LRU" : "Optimal");
    for (int i = 0; i < n; i++) {
        int hit = -1;
        for (int j = 0; j < nf; j++)
            if (frames[j] == pages[i])
                hit = j;
        if (hit != -1) {
            if (mode == 1)
                time[hit] = i;
        }
        else {
            int rep = -1;
            for (int j = 0; j < nf; j++)
                if (frames[j] == -1) {
                    rep = j;
                    break;
                }
            if (rep == -1) {
                if (mode == 0) {
                    rep = next;
                    next = (next + 1) % nf;
                }
                else {
                    int extreme = (mode == 1) ? i : -1;
                    for (int j = 0; j < nf; j++) {
                        int look = -1;
                        if (mode == 1)
                            look = time[j];
                        else {
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        for (int k = i + 1; k < n; k++)
            if (pages[k] == frames[j]) {
                look = k;
                break;
            }
        if (look == -1) {
            rep = j;
            break;
        }
    }
    if ((mode == 1 && look < extreme) ||
        (mode == 2 && look > extreme)) {
        extreme = look;
        rep = j;
    }
}
}
frames[rep] = pages[i];
if (mode == 1)
    time[rep] = i;
faults++;
}
printf("Page %d -> [", pages[i]);
for (int j = 0; j < nf && frames[j] != -1; j++)
    printf("%d%s", frames[j],
        (j < nf - 1 && frames[j + 1] != -1) ? " " : "");
printf("]\n");
}
printf("Total Page Faults (%s): %d\n",
    mode == 0 ? "FIFO" : mode == 1 ? "LRU" : "Optimal",
    faults);
}
int main() {
    int n, nf, *pages;
    printf("1WA24CS218\n");
    printf("Enter number of pages: ");
    scanf("%d", &n);
    pages = (int*)malloc(n * sizeof(int));
    printf("Enter page reference string:\n");

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for (int i = 0; i < n; i++)
    scanf("%d", &pages[i]);
printf("Enter number of frames: ");
scanf("%d", &nf);
for (int m = 0; m < 3; m++)
    simulate(pages, n, nf, m);
free(pages);
return 0;
}

```

```

1WA24CS218
Enter number of pages: 15
Enter page reference string:
7 0 1 2 0 3 0 4 2 3 0 3 1 2 0
Enter number of frames: 3

--- FIFO Page Replacement ---
Page 7 -> [7]
Page 0 -> [7 0]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 3 1]
Page 0 -> [2 3 0]
Page 4 -> [4 3 0]
Page 2 -> [4 2 0]
Page 3 -> [4 2 3]
Page 0 -> [0 2 3]
Page 3 -> [0 2 3]
Page 1 -> [0 1 3]
Page 2 -> [0 1 2]
Page 0 -> [0 1 2]
Total Page Faults (FIFO): 12

```

```
--- LRU Page Replacement ---
Page 7 -> [7]
Page 0 -> [7 0]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 0 3]
Page 0 -> [2 0 3]
Page 4 -> [4 0 3]
Page 2 -> [4 0 2]
Page 3 -> [4 3 2]
Page 0 -> [0 3 2]
Page 3 -> [0 3 2]
Page 1 -> [0 3 1]
Page 2 -> [2 3 1]
Page 0 -> [2 0 1]
Total Page Faults (LRU): 12

--- Optimal Page Replacement ---
Page 7 -> [7]
Page 0 -> [7 0]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 0 3]
Page 0 -> [2 0 3]
Page 4 -> [2 4 3]
Page 2 -> [2 4 3]
Page 3 -> [2 4 3]
Page 0 -> [2 0 3]
Page 3 -> [2 0 3]
Page 1 -> [2 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Total Page Faults (Optimal): 8
```